Pressure vessel stress & strain

EMAE 285 · Lab 2

Experiment overview

Pressurized a thin-walled aluminum cylinder and investigated hoop and longitudinal stress. Applied thin-wall assumptions and considered strain relationships and the influence of Poisson’s ratio.

Variables and instrumentation

Internal pressure, vessel dimensions, strain response

Analysis and presentation

Measured-versus-theoretical strain/stress; assumptions and error discussion

Engineering relevance

Directly relevant to pressure systems, energy equipment and validation of structural models.

Scope and attribution

Course laboratory / group work unless noted. The resonance experiment is a proposed design and has no claimed measured outcomes.

Portfolio case-study brief based on coursework and user-provided descriptions. This is NOT a copy of the original laboratory report. Diagrams on the website are explanatory illustrations, not photographs or original experimental plots.

Original source (may require Google Drive permission):
https://docs.google.com/document/d/1iv_cpZb6SSkqzYTCpfoBsTbkwef9caFw0AnThAmYZ7U/edit